Laser Wavelength Reference Guide

Every laser wavelength — chromophore targets, clinical applications, skin type safety, and device examples

ESTHETIC REFERENCE

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LASER WAVELENGTH REFERENCE — COMPLETE GUIDE

Wavelength	Laser Type	Primary Chromophore	Clinical Applications	Skin Type Safety
532nm (KTP/Nd:YAG doubled)	KTP / frequency-doubled Nd:YAG	Oxyhemoglobin, melanin	Superficial vascular (telangiectasias, rosacea, port wine), pigmented lesions	I–III only. High melanin competition in darker skin — burn risk
585–595nm	Pulsed Dye Laser (PDL)	Oxyhemoglobin	Vascular lesions, rosacea, PWS, scars, striae	I–III preferred. Purpura common. Most selective vascular laser available
694nm	Ruby laser	Melanin (high absorption)	Tattoo removal (black/dark inks), pigmented lesions, hair removal	I–II only. High melanin absorption = burn risk in darker skin
755nm	Alexandrite	Melanin (strong absorption)	Hair removal (light-medium skin), tattoo (black, blue, green inks), pigmented lesions	I–III. Test patch for Type III. Contraindicated IV–VI
810nm	Diode	Melanin (moderate)	Hair removal (versatile), vascular (with modifications)	I–IV. Most versatile for hair removal across skin types
1064nm	Nd:YAG	Melanin (lower), water, hemoglobin	Hair removal (dark skin), vascular (deeper), tattoo (black), skin tightening, onychomycosis	I–VI — safest wavelength for Fitzpatrick V–VI. Gold standard for dark skin
1320/1440nm	Non-ablative fractional Nd:YAG	Water	Collagen remodeling, skin tightening, mild resurfacing	I–VI with appropriate settings
1540/1550nm	Non-ablative fractional Er:Glass	Water	Acne scars, skin texture, fine lines, non-ablative resurfacing	I–V with conservative settings for darker skin
2940nm	Erbium:YAG (Er:YAG)	Water (highest absorption)	Ablative resurfacing, skin resurfacing — less thermal damage than CO2	I–III. Minimal thermal injury — fast healing vs CO2
10,600nm	CO2 laser	Water (very high)	Full ablative resurfacing, skin tightening, lesion removal	I–III. Gold standard ablative. Highest downtime. Greatest results

IPL (INTENSE PULSED LIGHT) — NOT A LASER

IPL Filter	Wavelength Range	Target	Clinical Use	Skin Type Safety
515/560nm filter	515–1200nm	Melanin + hemoglobin (broad)	Photo-rejuvenation, pigmentation, rosacea, hair removal	I–III. Caution III. Contraindicated IV–VI
590/615nm filter	590–1200nm	Hemoglobin (more selective)	Vascular lesions, rosacea, telangiectasias, diffuse redness	I–III. Safer for vascular-focused treatment
640/695nm filter	640–1200nm	Hair (melanin in follicle)	Hair removal — lighter skin	I–III for hair removal
Note	IPL is broadband light — not coherent like laser	—	Less targeted than laser — operator skill critical. Versatile but less precise	Always more conservative with IPL in darker skin types

CLINICAL APPLICATION QUICK REFERENCE

Hair removal — light skin	Alexandrite 755nm (gold standard) or diode 810nm. IPL also effective.
Hair removal — dark skin	Nd:YAG 1064nm ONLY. Safest for Fitzpatrick IV–VI. All other wavelengths carry burn risk.
Vascular lesions	PDL 585/595nm (most selective). KTP 532nm for superficial. Nd:YAG 1064nm for deeper vessels.
Pigmented lesions	Q-switched Nd:YAG or Picosecond for lentigines/PIH. Ruby 694nm for very superficial. IPL for photo-damage.
Tattoo removal	Picosecond lasers now gold standard. Multiple wavelengths needed for multicolor tattoos. Black: 1064nm. Red: 532nm. Green/blue: 755nm.
Skin resurfacing	CO_2 (10,600nm) for maximum results. Er:YAG for less downtime. Non-ablative fractional for no downtime option.
Skin tightening (non-ablative)	Nd:YAG 1064nm, non-ablative fractional, or RF-based technologies (not laser-based).

PRE-TREATMENT REQUIREMENTS — ALL LASER/IPL

❌ No active tan or self-tanner — 4 weeks minimum clearance required before ANY laser or IPL treatment

❌ Accutane within 6–12 months — impaired wound healing, increased scarring risk

❌ Photosensitizing medications — doxycycline, tetracycline, St. John's Wort, certain antibiotics (check full list)

✓ Always perform a test patch on a small area for new patients before treating a full area — especially Fitzpatrick III–IV

✓ Document Fitzpatrick type, settings used, and patient response for every treatment — adjustments guided by documented history